Answer Part 1

Engineering workflow and peer review

Peer-review candidate - not yet a completed submission. Released product tested on e1fa72f9fa1ad26ead74f82cf1de6bbf09de6384. Final-evidence approval/merge, final Kanban Done and submission acceptance remain human gates; no approval is invented.

Student: Phraewa Saphanont, 67070507213 (@PhraewaS). Peer: Natthawadee Phukhema, 67070507201 (@guluJa). Official Thai names and original quotations: Full reviewer.md.

Feature branches -> lab4-staging -> approved release #63 -> main e1fa72f -> codex/lab4-final-verification-evidence -> reviewed evidence PR -> main -> final acceptance. Product code is unchanged by this evidence branch.

Own PR / scope	Reviewed head / approval	Actual merge / branch target
#55 [Lab 4] Engineering Contract and Test Plan	0580fc1 APPROVED	ae41448 / lab4-staging
#56 [Lab 4] Actions Taken data model and backend API	eb9ae0a APPROVED	70f8c45 / lab4-staging
#57 [Lab 4] Ticket workflow and resolution enforcement	df882a1 APPROVED	5e974d3 / lab4-staging
#58 [Lab 4] Requester and Staff dashboards	6a07364 APPROVED	514f9b6 / lab4-staging
#59 [Lab 4] Actions Taken in Ticket Detail	b4e6315 APPROVED	c9b4f46 / lab4-staging
#60 Lab 4 verification and E2E evidence (#53)	2f980e0 APPROVED	2741c4b / lab4-staging
#61 [Lab 4] Staging integration and release readiness	201a8df APPROVED	b618547 / lab4-staging
#62 [Lab 4] Restore recovery verification with PostgreSQL SQL tools	53e1375 APPROVED	c239461 / lab4-staging
#63 [Lab 4] Release approved integration from lab4-staging to main	f95ba5c APPROVED	e1fa72f / main
#64 [Lab 4] Update Peer Review Records for Both Repositories	d4479bb APPROVED	5ab13bb / lab4-staging
#65 [Lab 4] Complete Release Documentation and Review Checklist	512f37c APPROVED	f95ba5c / lab4-staging

All #55-#60 features and #61/#62/#64/#65 maintenance were individually approved before staging merge. Release #63: formal approval 2026-10-14 14:38:54 UTC, merge 14:39:03 UTC. The contract #55 preceded completion of #56-#60.

Rendered reviewer record: both directions and original review receipt

Friend PR reviewed by @PhraewaS	Approval / actual merge
#68 [Lab 4] Engineering Contract and Test Plan	APPROVED f8598e5 / f4b28b8
#75 [Lab 4] Actions Taken Model, Migration, Seed and API Foundation	APPROVED 3525ef2 / af255b1
#76 [Lab 4] Actions Taken Ticket Detail UI	APPROVED 0e82dd5 / 284c7cb
#77 [Lab 4] Implement Ticket Workflow and Resolution Gate	APPROVED de4dcb6 / bb6049b
#78 [Lab 4] Implement Requester and Staff Dashboards	APPROVED bf04947 / 51bcf9c
#79 [Lab 4] Final Hardening, Regression, E2E and Visual Verification	APPROVED af6fe86 / 3b7291e
#80 [Lab 4] Pre-release Audit and Staging Verification	APPROVED f35b334 / 48694b8
#81 [Lab 4] Release integration into main	APPROVED 48694b8 / 08328cd
#82 [Lab 4] Final-main Verification and Final Documentation	APPROVED a5538c0 / fc67534
#83 [Lab 4] Integrate Final Documentation and Verification Evidence	APPROVED fc67534 / 99f9eb3

Current paginated snapshot: 21 PRs, 111 events (61 conversation comments, 50 submitted reviews, 0 inline comments). Every original quotation is preserved in the repository; approval is checked against the reviewed head and merge time. Historical snapshots remain unchanged.

ตรวจ release head f95ba5c แล้วนะคะ งานเอกสาร #65 รวมเข้ามาครบ และผล verification รอบใหม่ระบุ source ตรงกันทั้งหกข้อนะคะ ตรวจ JSON เทียบกับ console output แล้วจำนวน tests/skips, recovery, metrics และ performance สอดคล้องกัน ไม่มี Product changes เพิ่มจากงานที่ตรวจไว้ รอบนี้ไม่พบจุดที่ต้องแก้เพิ่มเติมก่อนอนุมัติ Release PR นี้ค่ะ

Original peer review; the empty-body APPROVED event is recorded as a state, not a fabricated quotation.

Author response - exact final paragraph excerpt: ขอบคุณนะคะที่ช่วยตรวจสอบให้กันมาตลอดเลยนะคะ

Complete original response including quoted review. Full rounds and corrective responses for all PRs remain rendered in reviewer.md.

Repository and Project evidence

README records setup, migration/seed, suite/build/E2E commands and six-phase main verification. .gitignore excludes .env, dependencies, builds, local attachment bytes and transient output.

Tracked tree: server/{src,tests,prisma}; client/{src,tests}; e2e/lab-04; docs/lab-04/{specification,tests,ui-spec,api-spec,reviewer,ai-use}.md; artifacts/lab-04/{screenshots,evidence}; scripts/.

Project / Kanban: #48-#53 Done; #54 open pending final acceptance. Capture all-Done only afterwards.

Answer Part 2

Rendered specification: requirements, rules and design

Full specification.md and Full api-spec.md. Scope: operational Actions under existing Tickets, resolution evidence gate, backend-calculated role dashboards; no additional Action assignment/status/delete workflow.

FR-01: A Ticket may have zero, one, or many Actions Taken.

FR-02: An Action Taken contains Action Date/Time, Action Description, Result, authenticated Performed By, Follow-Up Required, conditional Follow-Up Note, and optional Attachment Notes.

FR-03: IT Staff and Administrators may create and update Actions Taken on an accessible Ticket. Requesters may not mutate them.

FR-04: The backend derives **performedBy** from the session; clients cannot supply or override the performer field.

FR-05: Requesters can list Actions Taken only for their own Ticket; Staff and Administrators can list them for Staff-accessible Tickets.

FR-06: Action updates require the last **updatedAt** value; missing/invalid versions return **400 VALIDATION_ERROR**, and a stale version returns a safe **409 ACTION_UPDATE_CONFLICT** without overwriting the newer row.

FR-07: **RESOLVED** requires a persisted Action Taken. A Requester's "appears resolved" indication remains advisory and never changes the formal status.

FR-08: Requester Dashboard metrics are **openTickets**, **waitingForRequester**, **recentlyUpdated**, and **recentlyResolved**; each list item drills into the existing Ticket Detail.

FR-09: IT Staff Dashboard has four metrics: unassigned active Tickets, active Tickets owned by the current user, all-time Actions Taken by the current user, and urgent high-priority Tickets. Status/priority breakdowns and bounded recently updated Ticket, urgent Ticket, and recent Action lists are separate response fields.

FR-10: Dashboards calculate from authoritative database queries, return concise summaries, and return safe **500 INTERNAL_ERROR** responses on failure.

FR-11: All state-changing browser requests keep the existing session-cookie and Origin protections.

FR-12: Existing Lab 1-3 routes and regression behavior remain intact unless this contract explicitly adds the Lab 4 resolution gate or Actions Taken capability.

Business rules - rendered specification

BR-01: Every Action Taken belongs to exactly one Ticket.

BR-02: The Ticket Owner coordinates the Ticket, but an Action Taken may be performed by a different active IT Staff member or Administrator.

BR-03: **performedById** is the authenticated actor, never a body field.

BR-04: Action Description and Result contain 1-5,000 trimmed characters.

BR-05: If Follow-Up Required is true, Follow-Up Note is required; otherwise it is stored as null.

BR-06: Attachment Notes are optional and limited to 2,000 characters.

BR-07: Action list ordering is deterministic by Action Date/Time ascending, then Action ID ascending.

BR-08: The additive migration preserves all earlier rows and foreign keys. Seed is idempotent and demonstrates zero, one, and multiple Actions Taken.

BR-09: Open Dashboard Tickets are **NEW**, **OPEN**, **IN_PROGRESS**, **WAITING_FOR_REQUESTER**, or **REOPENED**; **CLOSED** and **CANCELLED** are not open.

BR-10: Recently updated/resolved Dashboard windows use the last 30 days in server time; counts and list data use the same authoritative query boundary.

BR-11: A Requester's resolved indication is advisory and does not set formal **RESOLVED** or **CLOSED** status.

BR-12: Staff status transitions remain the Lab 3 permitted transition matrix; a transition to **RESOLVED** additionally requires at least one Action Taken.

BR-13: Assignment is a parent-Ticket operation, not an Action field. Only IT Staff may perform Assignment; a non-null chosen Owner must be an active IT Staff or Administrator account. Eligibility as an Owner does not grant Administrators permission to perform Assignment or formal status writes. Administrators retain Ticket read/IT-Priority access and the expressly granted Action create/edit capability.

BR-14: Action Date/Time must be a string parseable as a date by the existing backend. Clients submit an ISO-8601 timestamp with an explicit offset/UTC; the response normalizes to UTC and storage retains millisecond precision. The existing handler has no past/future bound. Editing a different field preserves the exact stored time, including seconds.

BR-15: Action creates require an **Idempotency-Key** UUID. A unique stored key and normalized-request fingerprint bind the Ticket, actor and draft. An exact retry returns the same record; changed-request reuse returns a conflict. The key is not a substitute for update-version checking.

BR-16: Entered data survives recoverable refresh/save failures. A stale edit requires refresh; an uncertain create keeps the submitted draft/key stable until retry or deliberate discard followed by reconciliation. Server errors disclose no credentials, SQL, paths or stack traces.

Status transitions, calculations and database decisions

Current	Permitted next	Additional rule
NEW	OPEN	IT Staff only
OPEN	IN_PROGRESS / WAITING_FOR_REQUESTER / CANCELLED	IT Staff only
IN_PROGRESS / WAITING_FOR_REQUESTER	WAITING_FOR_REQUESTER or IN_PROGRESS / RESOLVED / CANCELLED	No no-op; RESOLVED requires a persisted Action
RESOLVED	CLOSED / REOPENED	Requester indication is advisory only
CLOSED	REOPENED	Administrator cannot perform formal status/assignment writes
REOPENED / CANCELLED	None	Unlisted/no-op transition: 409

Active = NEW, OPEN, IN_PROGRESS, WAITING_FOR_REQUESTER, REOPENED. Recent = inclusive updatedAt/actionDateTime >= one captured server UTC now minus 30 days; no upper bound. Requester counts are owned only. Staff unassigned/owned/urgent count active Tickets; myActionsTaken counts all actor Actions. Counts are not capped; recent lists are capped at 10 with deterministic time/id tie-breaks.

Normalized ActionTaken has restricted Ticket/actor FKs, TIMESTAMPTZ(3), description/result VARCHAR(5000), conditional note VARCHAR(5000), optional attachment note VARCHAR(2000), createdAt/updatedAt, nullable legacy UUID create key and SHA-256 fingerprint. Chronology indexes plus unique key. Two additive migrations create the child table and retry-key columns; no historical Ticket/action fabrication.

Design decisions: normalized child separates Owner from performer; restricted FKs preserve history; mandatory update token prevents stale writes; unique key+fingerprint makes create retry safe; authoritative queries avoid duplicate/cached aggregates. Seed demonstrates zero, one and multiple Actions and is repeatable. Recovery is isolated, not destructive rollback on user data.

Acceptance criteria and Product Definition of Done

AC-01: Given an authenticated Staff/Admin, an existing Ticket and a valid draft/key, when an Action is created, then return 201 with the correct Ticket, immutable actor and documented fields.

AC-02: Given an invalid field/ID, absent resource, unauthorized role/ownership or backend failure, when the route is called, then return the documented safe error without exposing protected data or saving invalid input.

AC-03: Given a loaded Action, when the full edit draft and current **updatedAt** are sent, then update it; missing/invalid tokens return 400, and stale tokens return 409 without overwriting the current row.

AC-04: Given a permitted Staff transition, when resolution is requested without an Action, then reject with 409; with an Action allow resolution and closure. Stale loaded status and read/write races conflict; Requester advisory remains non-formal; cancellation and inactive-assignee rejection retain Lab 3 rules.

AC-05: Given each authenticated role, when its Dashboard loads, then counts, bounded lists, ownership, zero states and first-request drill-down filters match the backend query contract, and selected counts match independent DB/API/UI evidence.

AC-06: Given desktop/tablet/mobile widths, when Dashboard/Actions controls are used, then responsive layouts fit horizontally, required/readonly fields are distinguishable, keyboard focus/activation works and Zen Green visual conventions remain consistent. Loading/empty/forbidden/failure states remain distinct.

AC-07: Given the guarded configured environment, when the complete regression/verification is run, then suites/builds, additive migration, seed, recovery, performance smoke and legacy role/navigation/attachment/communication/user-management checks pass with intentional skips recorded separately. Final completion requires a repeat on released main.

AC-08: Given an unsaved/uncertain draft or a precise stored time, when refresh/retry or a different-field edit occurs, then preserve entered data until deliberate discard, reuse the create key without duplicates, reconcile after uncertain discard and retain the unchanged exact Action Date/Time.

Executed product gates: six documents, additive migration/seed/recovery, full suites/builds/E2E, role/error/stale/retry cases, metric comparison, responsive focus/visual capture on released main. Submission gates still require human final-evidence approval, accepted PDF and final Project Done. Specification history: #55 approval.

Answer Part 3

Rendered test plan, traceability and actual main output

Full tests.md; complete phase JSON and raw output. All six phases ran sequentially on e1fa72f9fa1ad26ead74f82cf1de6bbf09de6384; localhost PostgreSQL 5433, isolated guarded E2E databases, credentials omitted.

Main check	Actual result	Coverage / limitation
Recovery safety	6 passed	Command/target/tool-version safety
Prepare / recovery	5 migrations; 9 tables + 5 migration-history rows matched	Temporary database and dump removed; physical attachment bytes not included
Server	164 passed; build exit 0	Unit/API/integration/auth/roles/workflow/regression/performance
Client	89 passed; build exit 0	Component/UI/style/accessibility/navigation/draft/retry
Lab 4 Playwright	11 passed / 10 intentional skips	21 total; 0 failed/flaky; mutation once on desktop, separate responsive suite
Lab 3 Playwright	26 passed / 16 intentional skips	42 total; 0 failed/flaky; complete authenticated legacy regression
Selected metrics	8 DB = API = UI values	Guarded exact same DB URL; independent query client
Keyboard / browser	6 responsive flows passed	Tab/Enter/Space/focus-visible, fit; 0 unexpected HTTP/page errors
Performance	40 requests / 0 failures	10 per role-dashboard/action endpoint; all maxima <1000 ms

Actual endpoint maxima (ms): 360, 14, 11, 8. This is local smoke evidence, not production load testing or a GitHub CI check.

Executed commands

node --test scripts/postgres-recovery.test.mjs; then node scripts/verify-lab4.mjs prepare|recovery|server|client|lab4|lab3 --env-file [existing untracked file] --output-dir artifacts/lab-04/evidence/final-main. Exact individual commands and exits are retained in summary.md and phase logs; no secret is required in this PDF.

All eight ACs: expected outcome, real paths, final product status

AC / test IDs	Expected result and actual test paths	Final product status
01 / UNIT-04, API-04, UI-04, E2E-04	Create/list/edit correct actor and fields; server/tests/lab-04/actions-taken.api.test.ts (actions.api.test.ts implementation), client/tests/lab-04/ActionsTaken.test.tsx, e2e/lab-04/actions-taken-flow.spec.ts	Passed on e1fa72f; submission acceptance pending
02 / API-04/05/06, REG-04	Safe body/ID/resource/role/ownership errors and inactive Owner rejection; same Actions tests plus server/tests/lab-03/staff-operations.integration.api.test.ts and users-admin.api.test.ts	Passed on e1fa72f; submission acceptance pending
03 / API-04, UI-04	Missing/invalid updatedAt 400, stale 409 without overwrite; Actions API/component paths above	Passed on e1fa72f; submission acceptance pending
04 / API-06, UI-06, E2E-07	Permitted matrix, resolution evidence, close/cancel and concurrent preconditions; server/tests/lab-04/ticket-workflow.api.test.ts, client/tests/lab-04/TicketWorkflow.test.tsx, e2e/lab-04/ticket-resolution.spec.ts	Passed on e1fa72f; submission acceptance pending
05 / API-05, UI-05, E2E-05/06, PERF-04	Counts/query boundaries/filters and bounded lists; server/tests/lab-04/{requester,staff}-dashboard.api.test.ts; client/tests/lab-04/{Requester,Staff}Dashboard.test.tsx; e2e/lab-04/dashboards.spec.ts; performance-smoke.test.ts	Passed on e1fa72f; submission acceptance pending
06 / RESP-04, STYLE-04	Three widths, labels/readonly/focus/no overflow; e2e/lab-04/responsive.spec.ts plus final visual harness/gallery; manual image inspection supplements, not pixel-diff/axe certification	Passed on e1fa72f; submission acceptance pending
07 / REG-04, E2E-00, RECOVERY-04	Migration/seed/recovery and all previous features; e2e/lab-04/migration-seed.spec.ts; scripts/postgres-recovery.test.mjs; server/client tests lab-01..03 and e2e/lab-03	Passed on e1fa72f; submission acceptance pending
08 / API-04, UI-04	Draft retention, safe exact retries/races, changed-key conflict and unchanged seconds; Actions API/component paths above	Passed on e1fa72f; submission acceptance pending

Planned expectations and historical feature outcomes are preserved in tests.md. Passing means the named assertion/method ran; deterministic append-only routes and Action tie-break ordering also have explicitly labelled source inspection. Intentional skips are not converted into passes.

Actual console extracts from main

server-tests.txt: Test Files 28 passed (28) | Tests 164 passed (164)

client-tests.txt: Test Files 20 passed (20) | Tests 89 passed (89)

lab4-e2e.txt: 10 skipped | 11 passed (2.5m)

lab3-e2e.txt: 16 skipped | 26 passed (4.5m)

Answer Part 4

Rendered AI use and My Reflection

Full ai-use.md. Model: OpenAI Codex, GPT-5-based coding agent in the Codex desktop workspace.

1. Continue TokTickIT from the final Lab 3 repository state and follow the attached Lab 4 handout strictly.

2. Design an additive PostgreSQL/Prisma ActionTaken model with parent-child Ticket ownership, audit fields, indexes, seed behavior, and stale-update handling.

3. Define exact Lab 4 API response shapes, validation, authorization, conflicts, and safe errors.

4. Implement backend-calculated Requester and IT Staff dashboards with concise metrics, empty behavior, and drill-down IDs.

5. Extend the existing Staff Ticket Detail and Requester Ticket Detail with role-appropriate Actions Taken UI while preserving Zen Green accessibility and responsive conventions.

6. Build a Test DD that maps each requirement to unit, API, UI, responsive, E2E, migration, and regression coverage.

7. Review the implementation for accidental Lab 3 regressions, unknown fields, client-controlled performer identity, and stale overwrites.

8. Verify build/test output, diff hygiene, migration behavior, and final evidence against the handout's nine answer parts.

9. For Issue #53, prepare a separately guarded Lab 4 database workflow, migration/seed repeatability check, role-based E2E scenarios, responsive screenshots, and a bounded performance-smoke test; report only checks that actually ran.

10. Integrate Issue #54 using the established peer-review workflow, verify isolated backup/restore on port 5433, rerun actual suites, correct evidence and keep release/final-main/PDF gates pending until approved.

My Reflection - original document text

The specification-agent work was useful for turning the handout's examples into numbered requirements and making the ambiguous "recent" dashboard phrase testable as a 30-day server-side window. The coding-agent work was most valuable when it kept the Action Taken write authority in the backend, made the performer session-derived, and used an additive migration rather than rewriting Lab 2/3 data. I treated generated suggestions as proposals: I retained existing Lab 3 role boundaries for assignment/status/comments/notes, added only the explicitly approved Lab 4 Action Taken capability for Administrators, and used regression tests/builds as the final authority.

Evidence workflow note: suggestions were checked against the handout, product sources, original GitHub peer comments and actual main runs. No agent-generated approval replaces human review.

Answer Part 5

IT Staff Dashboard: accurate counts and drill-down

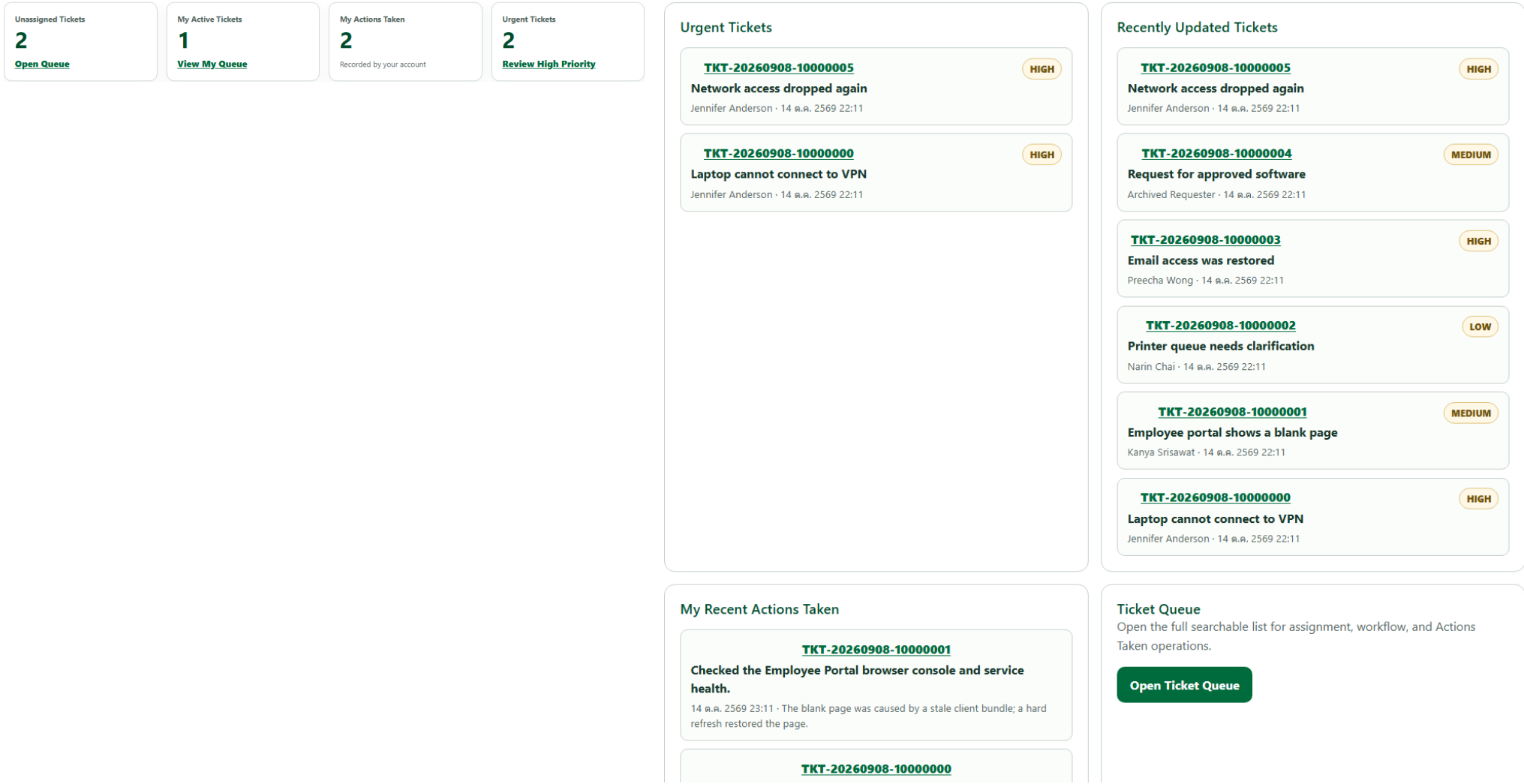
Selected metric	DB	API	UI
unassignedActive	2	2	2
myActive	1	1	1
myActionsTaken	2	2	2
urgentTickets	2	2	2

Unassigned -> ownerId=unassigned & activeOnly=true; My Active -> authenticated owner & activeOnly=true; Urgent -> HIGH & activeOnly=true. Breakdown status/priority buttons apply the exact filter across all statuses. Recent Ticket/Action buttons open Staff Ticket Detail by Ticket ID. Counts are authoritative and not the length of bounded arrays.

Dashboard states are visually distinct at every width. Success images use real seeded API data; zero/forbidden/failure/loading fixtures exercise the released UI with controlled HTTP replies and do not pretend to be live permission failures. Full API/component tests cover those backend outcomes.

IT Staff Dashboard - desktop metrics and recent operational lists

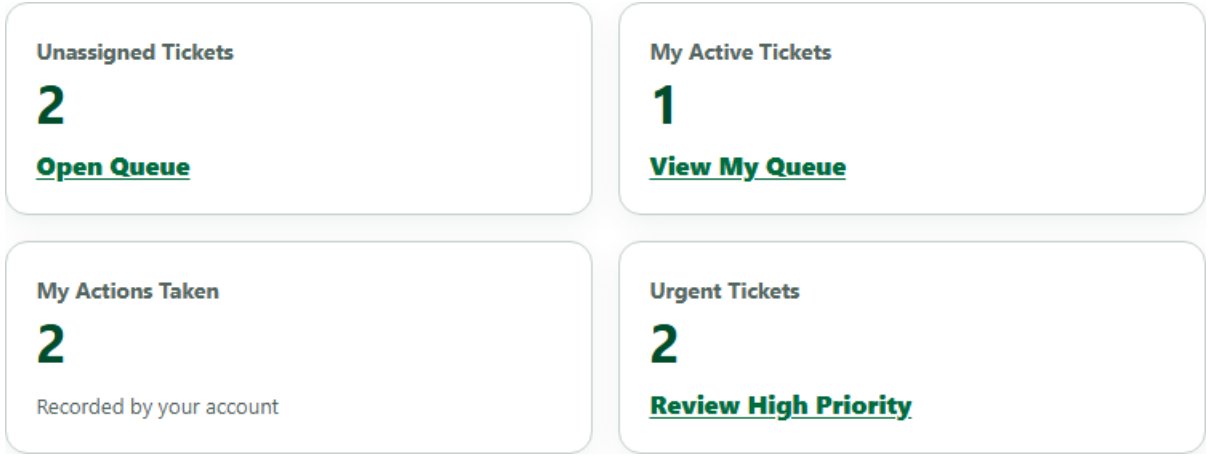
Actual released product e1fa72f. Full-page originals, state provenance and horizontal-fit checks: visual evidence gallery.

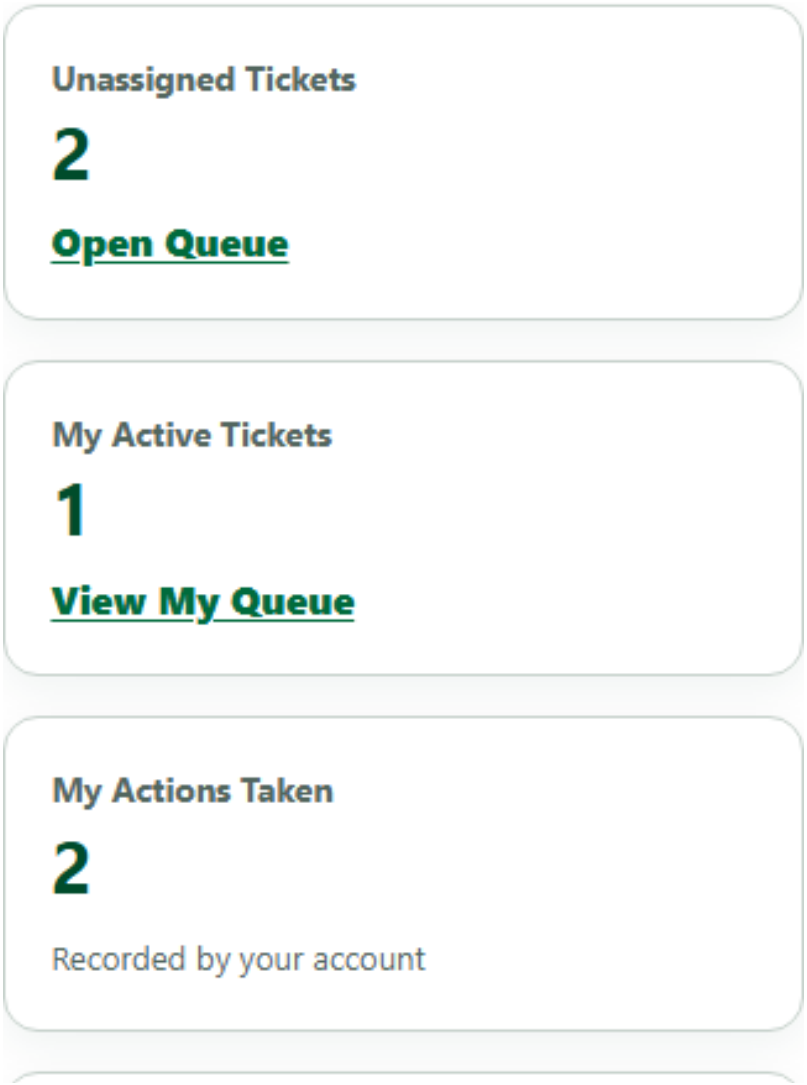


Selected readable image regions; full originals are linked in the gallery. desktop-staff-dashboard-metrics.png / desktop-staff-dashboard-lists.png

IT Staff Dashboard - tablet and mobile metrics

Actual released product e1fa72f. Full-page originals, state provenance and horizontal-fit checks: visual evidence gallery.





Selected readable image regions; full originals are linked in the gallery. tablet-staff-dashboard-metrics.png / mobile-staff-dashboard-metrics.png

Full-page breakdown/recent/urgent/My Recent Actions images: desktop-staff-dashboard-full.png, tablet-staff-dashboard-full.png, mobile-staff-dashboard-full.png.

Answer Part 6

Actions Taken and parent Ticket lifecycle

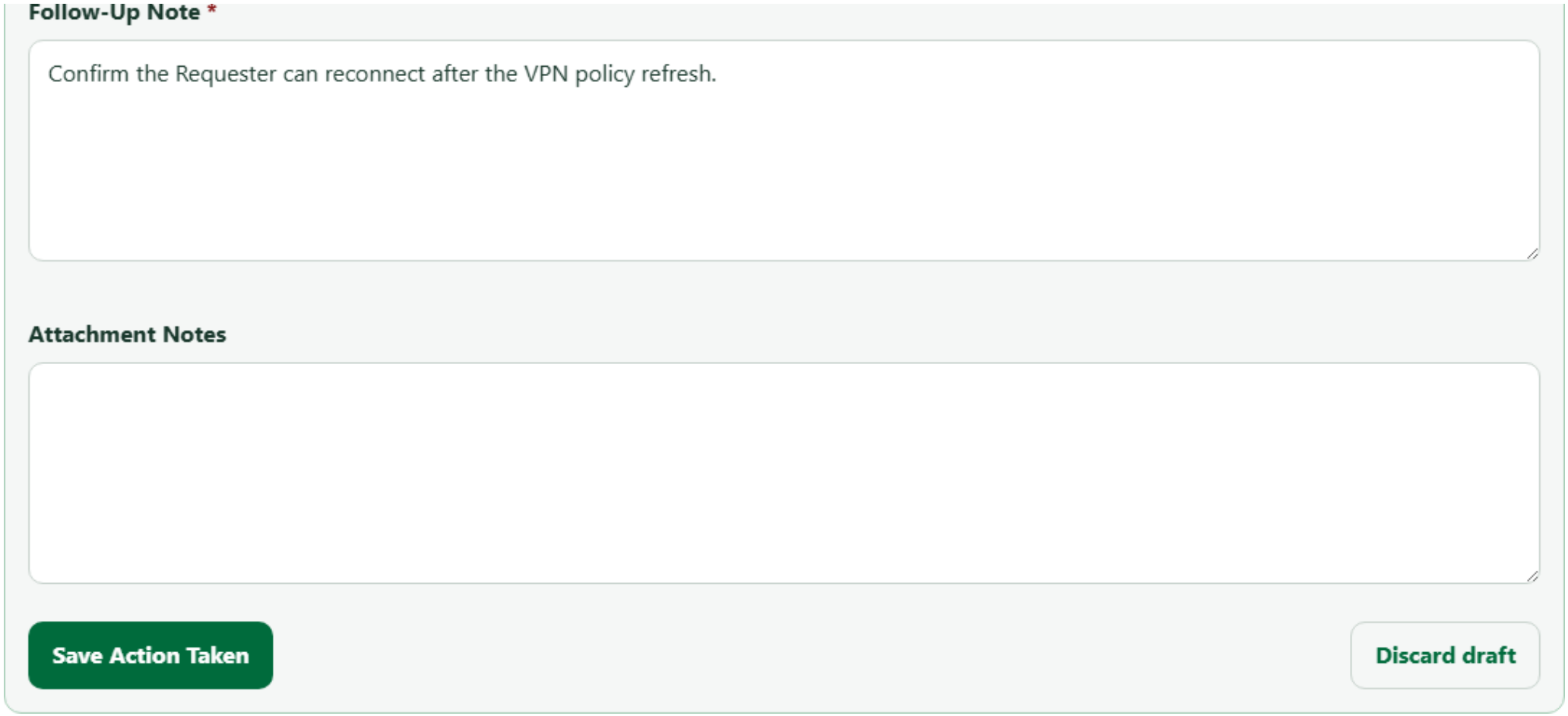
Actions Taken: list/create/edit, authenticated readonly actor, exact datetime preservation, conditional Follow-Up Note and optional Attachment Notes. A Ticket demonstrates multiple Actions. Requester sees only owned read-only history; Staff/Admin writes are backend-authorized.

Assignment, formal status, complete and cancel apply to the parent Ticket, as approved in #55/#63/#65. Only IT Staff assigns/transitions; active Staff or Administrator may be chosen as Owner. API/regression verifies inactive-assignee rejection. E2E-07 proves evidence-required resolution, RESOLVED -> CLOSED and a separate OPEN -> CANCELLED.

Versioned updates conflict safely; uncertain create reuses UUID/fingerprint; refresh/retry preserves draft until explicit discard/reconciliation. Actual 400 validation capture retains draft; controlled 409/500 visuals separately show conflict and failed-refresh retention. Backend/API/component suites provide actual stale/retry/role guarantees.

Actions Taken - desktop create: readable Follow-Up Note and Save controls

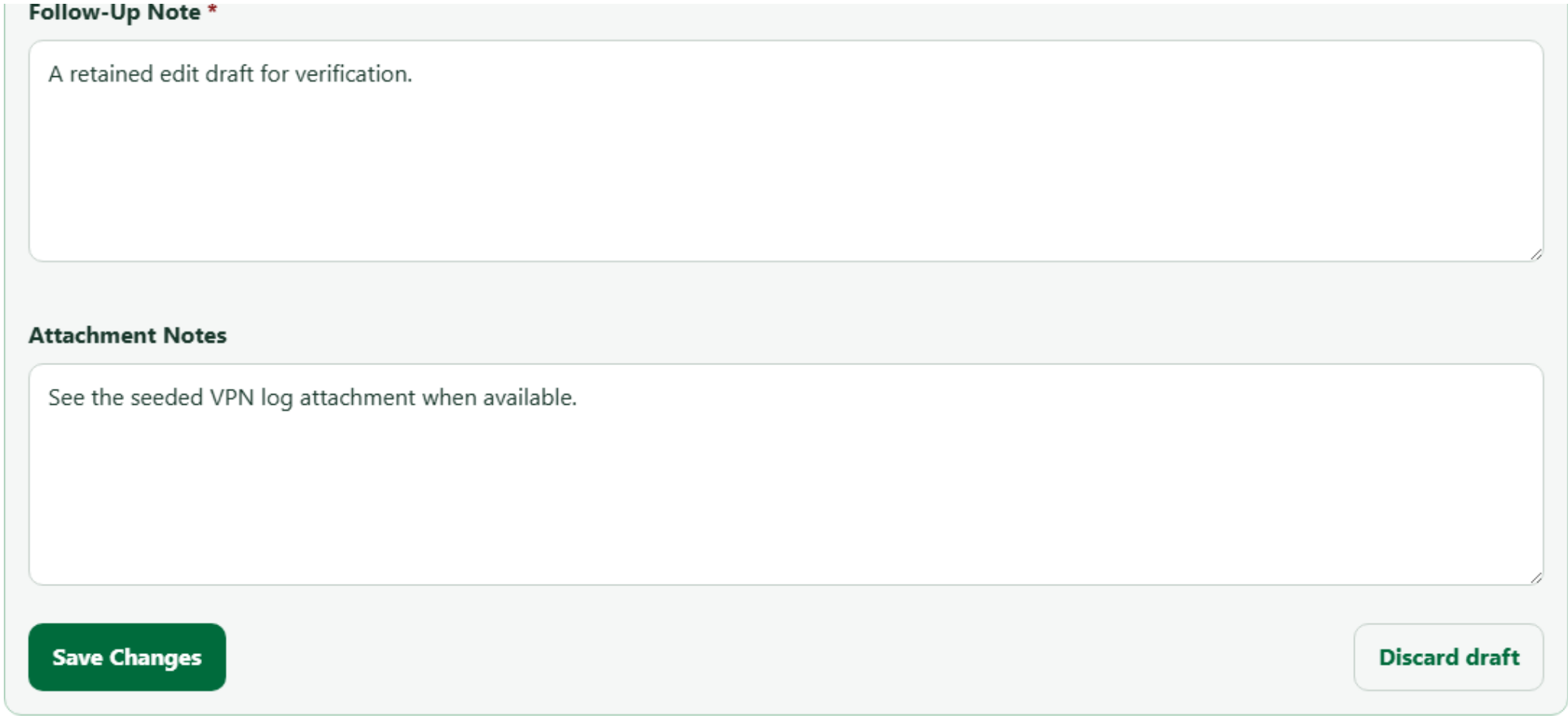
Actual released product e1fa72f. Full-page originals, state provenance and horizontal-fit checks: visual evidence gallery.



Selected readable image regions; full originals are linked in the gallery. desktop-actions-create-follow-up.png

Actions Taken - desktop edit: readable Follow-Up Note and Save Changes

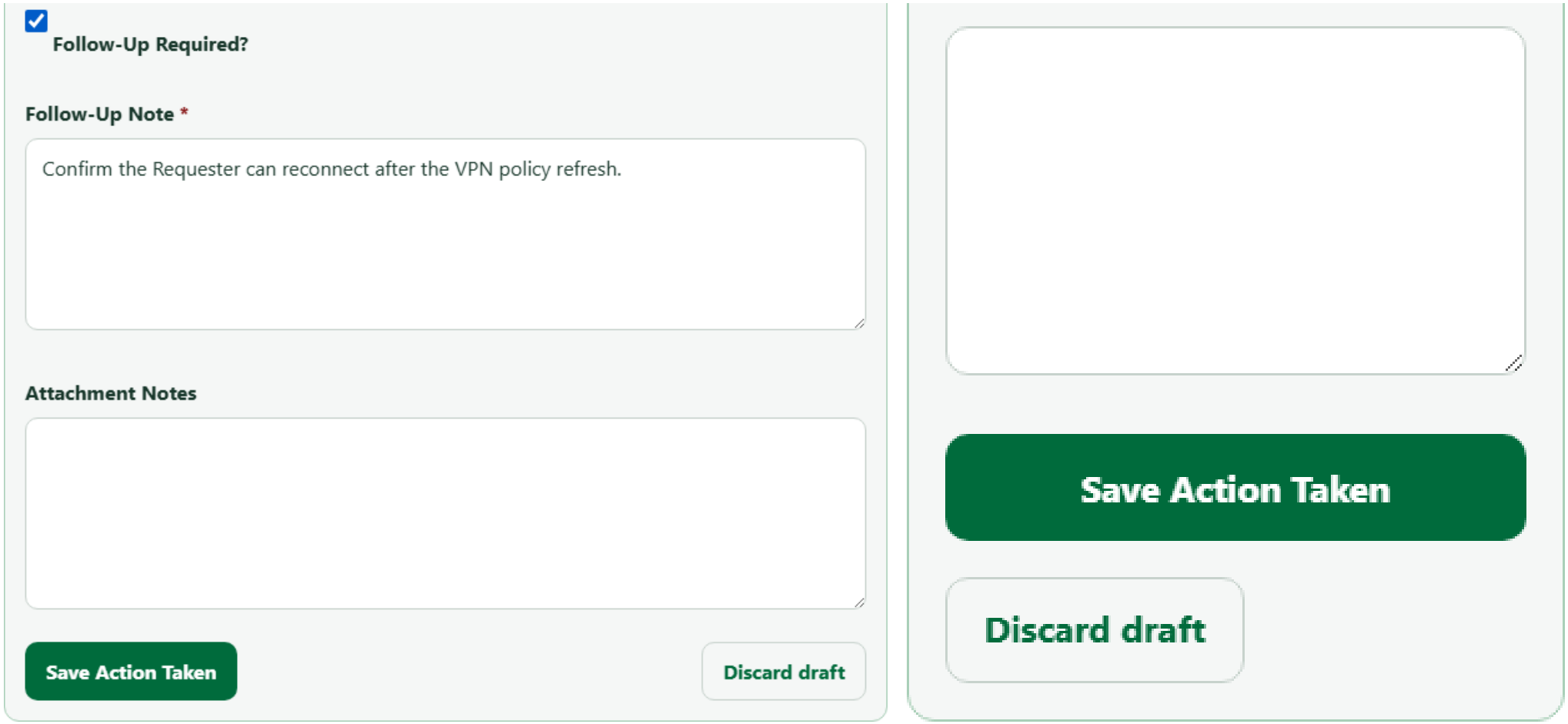
Actual released product e1fa72f. Full-page originals, state provenance and horizontal-fit checks: visual evidence gallery.



Selected readable image regions; full originals are linked in the gallery. desktop-actions-edit-follow-up.png

Actions Taken - tablet and mobile create editor regions

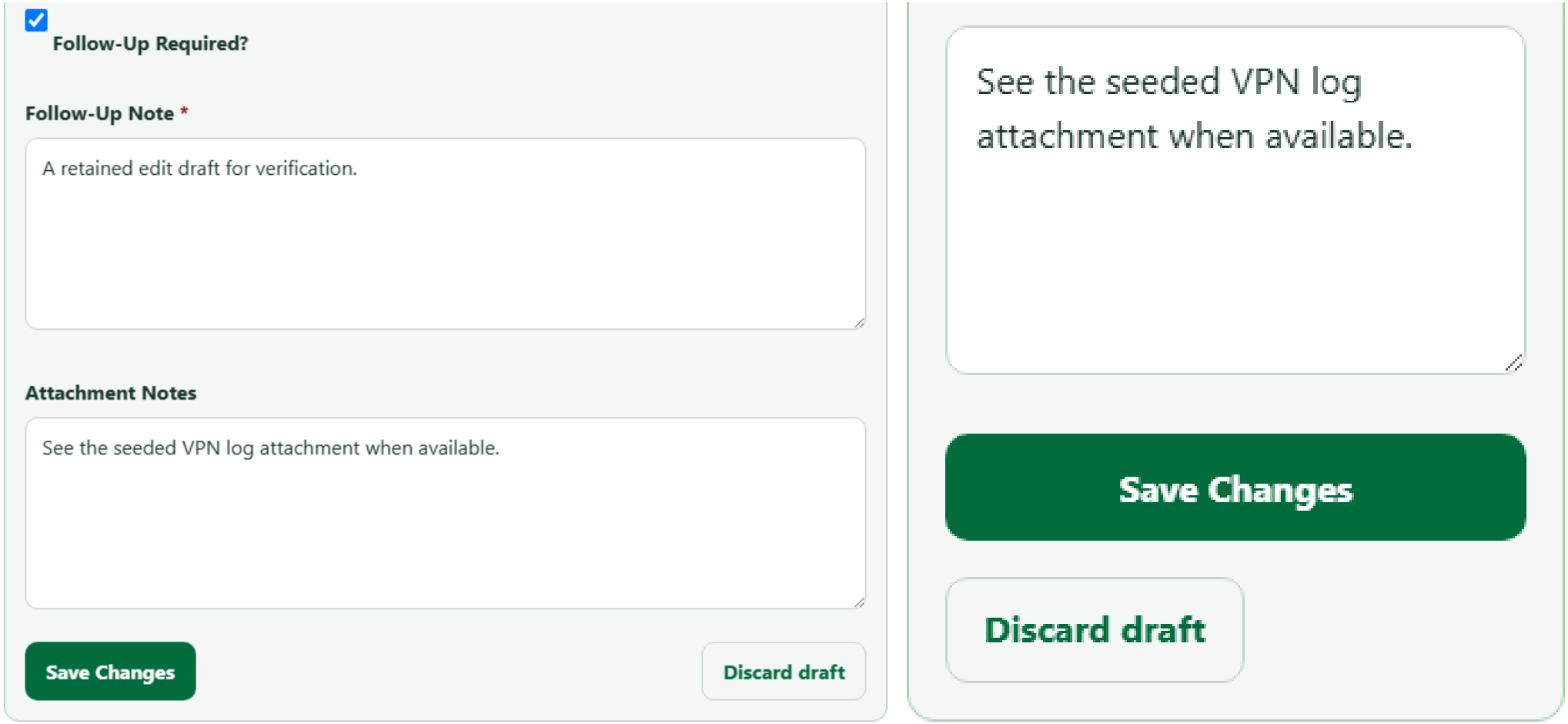
Actual released product e1fa72f. Full-page originals, state provenance and horizontal-fit checks: visual evidence gallery.



Selected readable image regions; full originals are linked in the gallery. tablet-actions-create-follow-up.png / mobile-actions-create-follow-up.png

Actions Taken - tablet and mobile edit editor regions

Actual released product e1fa72f. Full-page originals, state provenance and horizontal-fit checks: visual evidence gallery.



Selected readable image regions; full originals are linked in the gallery. tablet-actions-edit-follow-up.png / mobile-actions-edit-follow-up.png

Validation/conflict/failed-refresh/forbidden are captured at every width; all images and sources are listed in visual/results.json. Readonly actor remains inside viewport; a single-line value may scroll internally, not overflow the page.

Answer Part 7

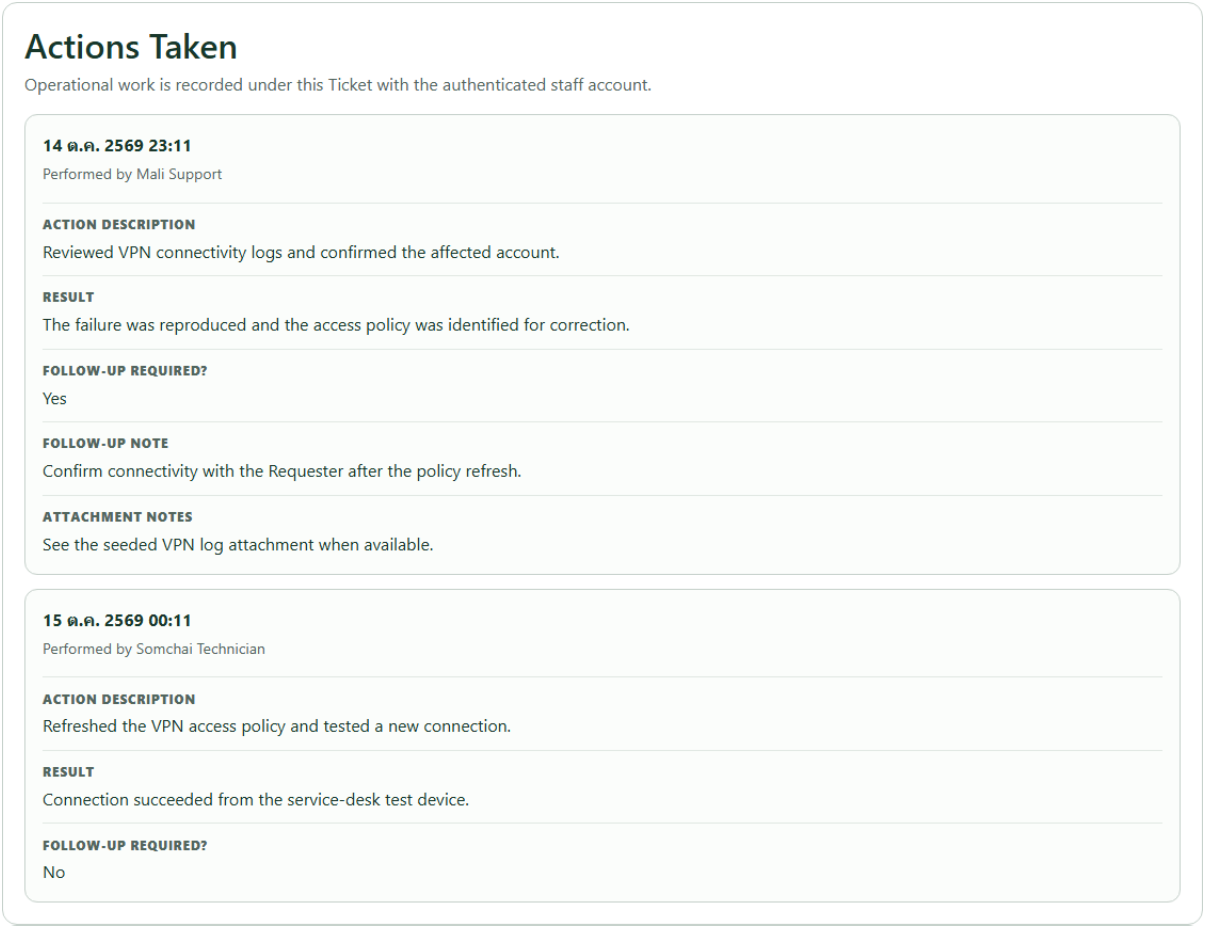
Ticket workflow, ordering and communication visibility

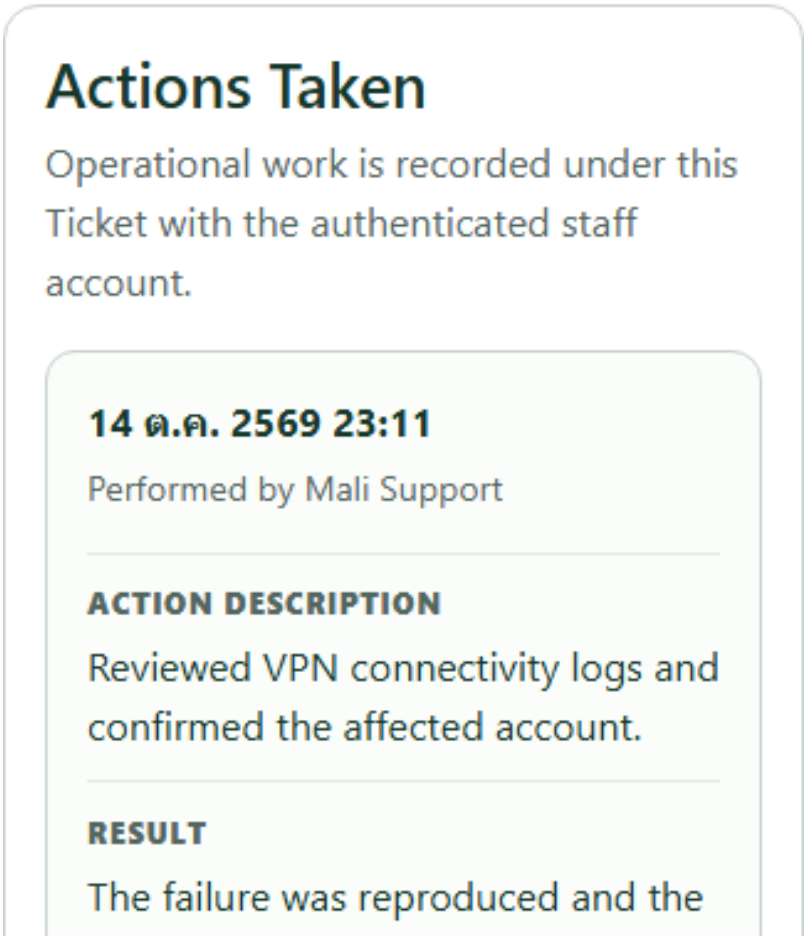
Acceptance	Evidence / outcome
Resolution prerequisite	e2e/lab-04/ticket-resolution.spec.ts: reject without Action; record work; resolve/close; cancel separately
Permitted matrix / role	server/tests/lab-04/ticket-workflow.api.test.ts and legacy Staff routes; no new Administrator status/assignment right
Stale page and read/write race	Client sends expectedCurrentStatus; server conditional write rejects conflict; Refresh Ticket required
Deterministic order	Dashboard/Queue assertions use time/id tie-breaks; Action query actionDateTime ASC,id ASC inspected in server/src/actions.ts
Append-only communication	Public Comments and Internal Notes list/create routes; no edit/delete; browser/API regression plus source route audit
Role visibility	Requester owned Ticket/Action/Public Comment only; Internal Notes staff-private; Administrator boundaries preserved; Actions explicitly editable with version

Ordering/append-only source inspection is not relabelled as a new automated case. Original main product sources: actions.ts, staff.ts, app.ts.

Requester read-only Actions - desktop and mobile regions

Actual released product e1fa72f. Full-page originals, state provenance and horizontal-fit checks: visual evidence gallery.





Selected readable image regions; full originals are linked in the gallery. desktop-requester-actions-panel.png / mobile-requester-actions-panel.png

Tablet read-only capture: tablet-requester-actions-panel.png. No Add/Edit controls on the Requester page; list values and applicable Follow-Up/Attachment Notes remain visible.

Answer Part 8

Requester Dashboard and final legacy regression

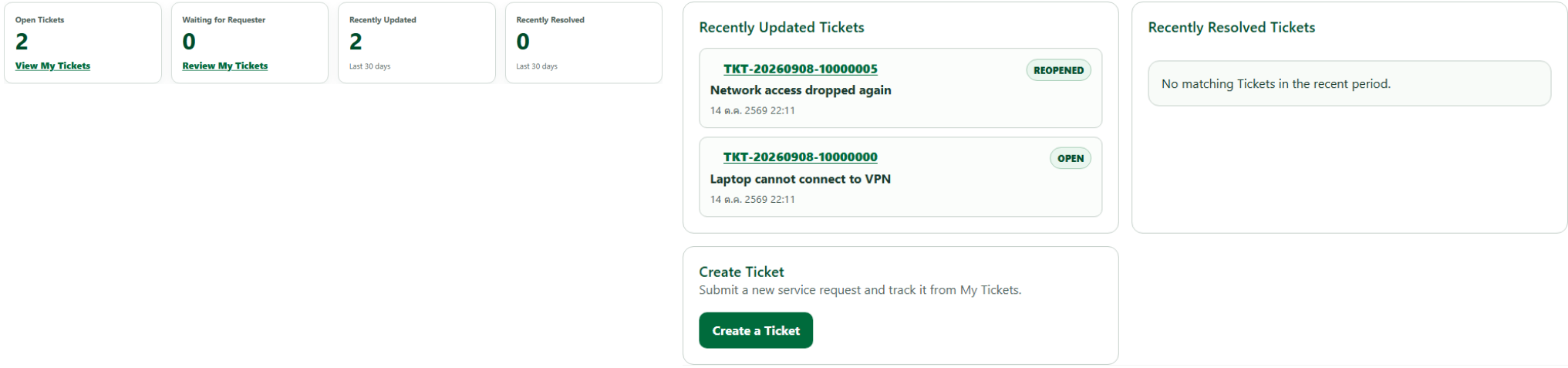
Selected metric	DB	API	UI
openTickets	2	2	2
waitingForRequester	0	0	0
recentlyUpdated	2	2	2
recentlyResolved	0	0	0

Open Tickets -> My Tickets ACTIVE filter; Waiting for Requester -> exact WAITING_FOR_REQUESTER filter. Recent Ticket buttons -> owned Detail. Recently updated/resolved counts are informational; no fabricated recent-window list filter. Zero lists have explicit text and remain distinct from failure.

Lab 3 main regression: 26 passed / 16 intentional viewport skips, 0 failed/flaky. Preserves login/logout/change-password, inactive/revoked access, My Tickets filters/sort/pagination, Create/Detail, attachment upload/download/soft-removal, Public Comments, Staff queue/assignment/priority/status, Internal Notes and Administrator users/edit/activation/password reset. Full Server/Client suites also cover role/ownership denials.

Requester Dashboard - desktop metrics and recent/empty lists

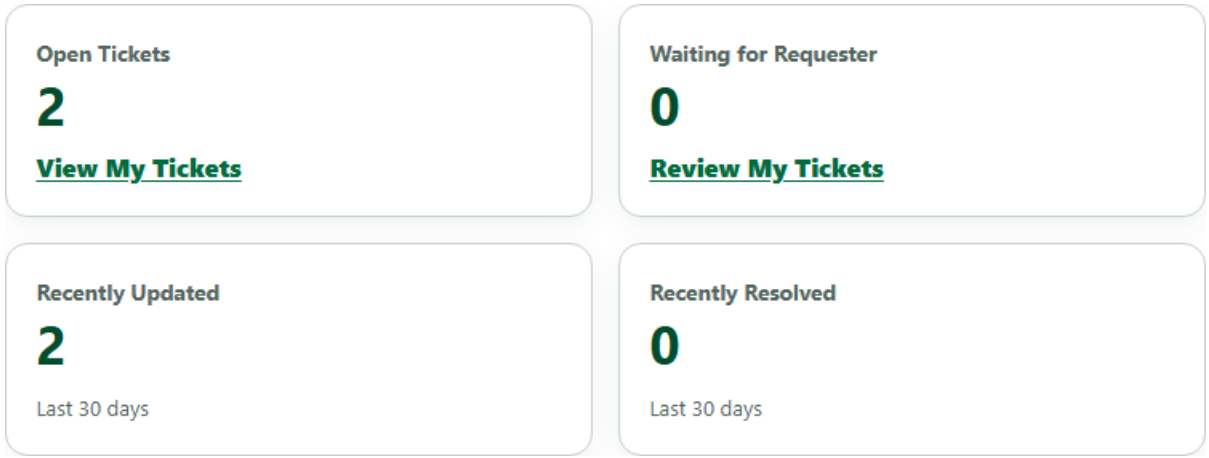
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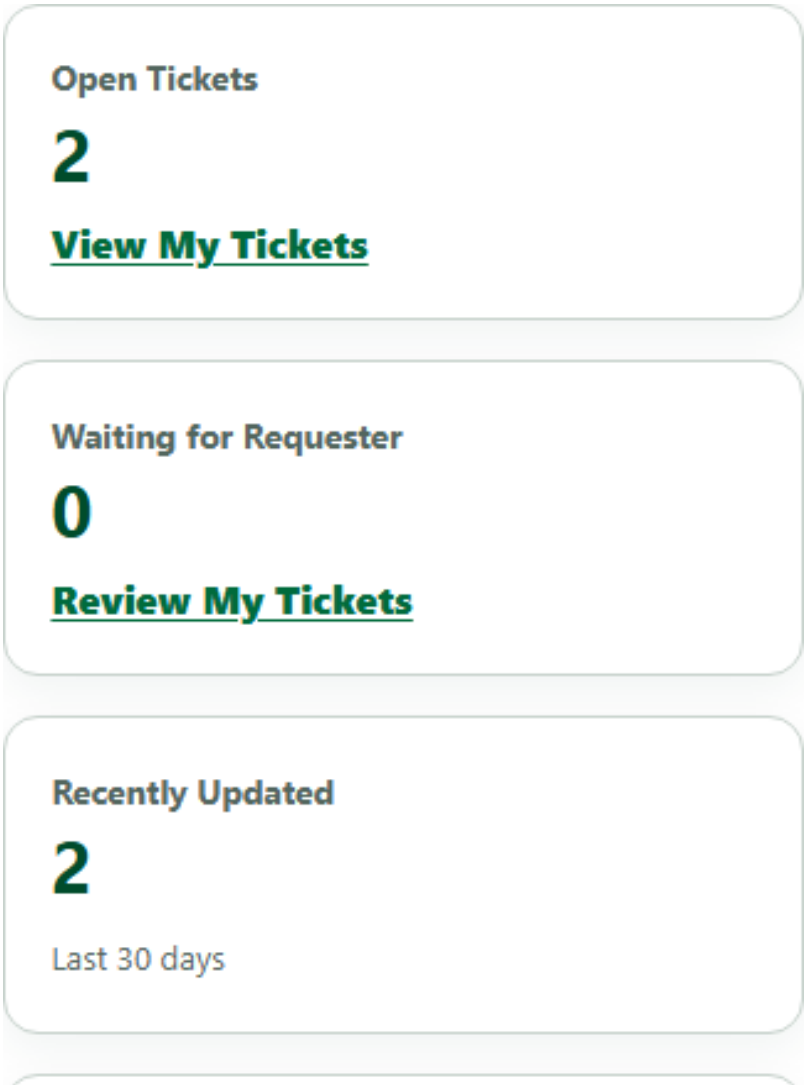


Selected readable image regions; full originals are linked in the gallery. desktop-requester-dashboard-metrics.png / desktop-requester-dashboard-lists.png

Requester Dashboard - tablet and mobile metrics

Actual released product e1fa72f. Full-page originals, state provenance and horizontal-fit checks: visual evidence gallery.





Selected readable image regions; full originals are linked in the gallery. tablet-requester-dashboard-metrics.png / mobile-requester-dashboard-metrics.png

Full-page and legacy screen originals remain in final-main/visual and final-main/lab3-screenshots; fresh sanitized Lab 3 HTML report is separate from historical reports.

Answer Part 9

Rendered UI specification and visual/accessibility acceptance

Full ui-spec.md and Full responsive-accessibility-checklist.md.

Every authenticated role sees an active **Dashboard** navigation item. Requesters retain My Tickets/Create Ticket. IT Staff and Administrators retain Ticket Queue; Administrators retain User Management. Active-page indication remains visible and the brand returns to the role dashboard.

Metric cards: Open Tickets, Waiting for Requester, Recently Updated, Recently Resolved. Recent list items show ticket number, summary, status, and updated time; each ticket number opens the existing read-only Ticket Detail. Empty lists state that no recent records match. A Create Ticket quick action remains available.

Metric cards: Unassigned Tickets, My Active Tickets, My Actions Taken, and Urgent Tickets. Breakdown cards show status and IT Priority counts. Lists show urgent and recently updated Tickets plus the current user's recent Actions Taken. Every actionable ticket opens Staff Ticket Detail or Queue. Administrator retains the Lab 3 administrative navigation and may create/update Actions Taken.

Requester and IT Staff/Administrator dashboards must specify these states independently for metric cards and each list:

Loading: show skeleton/placeholder content with an accessible **aria-busy** or live status; do not present stale values as current.

Success with data: show the calculated metric/list, timestamp where relevant, and the documented Ticket Detail, Queue, or Staff Ticket Detail drill-down.

Success with no records: show zero for the metric and a clear "No matching records" empty state for the affected list. Empty is not an API failure.

Failure: show a safe, non-technical message for **500 INTERNAL_ERROR** or network failure, preserve the dashboard shell, and provide a visible **Retry** action.

Forbidden: show a role-appropriate access message for **403 ROLE_FORBIDDEN** without rendering protected metrics or list data; provide a link back to an allowed dashboard or screen.

Retry/refresh: disable or label the retry control while the request is pending, then return to Loading, Success, Empty, or Failure based on the new response without duplicating requests.

These states must remain distinguishable without color alone and must work at desktop, tablet, and mobile widths.

Ticket Detail shows an ordered list with Action Date/Time, Description, Result, Performed By, Follow-Up Required, Follow-Up Note, and Attachment Notes. Staff/Admin see Add Action Taken and Edit controls. The form labels every field, marks required fields, reveals Follow-Up Note only when required, and shows saving/validation/conflict feedback. On a stale-edit conflict, Refresh preserves the draft through loading and failed retries; the user must explicitly choose whether to keep or discard it. If a create response is uncertain, show that the result may be unknown, lock the submitted fields, and allow retry with the same idempotency key; do not create another key unless the user explicitly discards that draft. After such a discard, reconcile the Action list before allowing another create. An unchanged Action Date/Time retains the precise stored timestamp, including seconds. Requesters see entries read-only and never see mutation controls.

IT Staff status controls show only transitions allowed by the status matrix. If a resolution attempt returns **409 RESOLUTION_ACTION_REQUIRED**, keep the formal status unchanged, reset the selector to its neutral prompt, and show a safe message explaining that an Action Taken must be recorded first. If a concurrent Staff status change returns **409 STATUS_UPDATE_CONFLICT**, keep the stale detail unchanged, show the safe conflict message, and provide a **Refresh Ticket** action before the Staff member retries.

All controls have semantic labels, visible focus, keyboard operation, live loading/status/error messages, and non-color status cues. Dashboard cards become one column on mobile. Ticket/action lists become stacked cards at narrow widths. Text wraps safely; page-level horizontal overflow, clipped controls, overlapping dialogs, and color-only status meaning are prohibited.

Fields, Zen Green rules and completed inspection scope

Reuse styles.css: green-950 **#004c2b**, green-800 **#006b3c**, green-700 **#0b7a46**, white surface, muted **#52665e**, border **#c7d2cc**, danger **#8a1c1c**. Metric/card hierarchy, rounded borders, shared primary/secondary/tertiary buttons, semantic status/priority text and labelled form sections match the earlier shell. Color complements text; it is never the only status or error cue. These tokens document the implemented theme, not a newly measured contrast ratio.

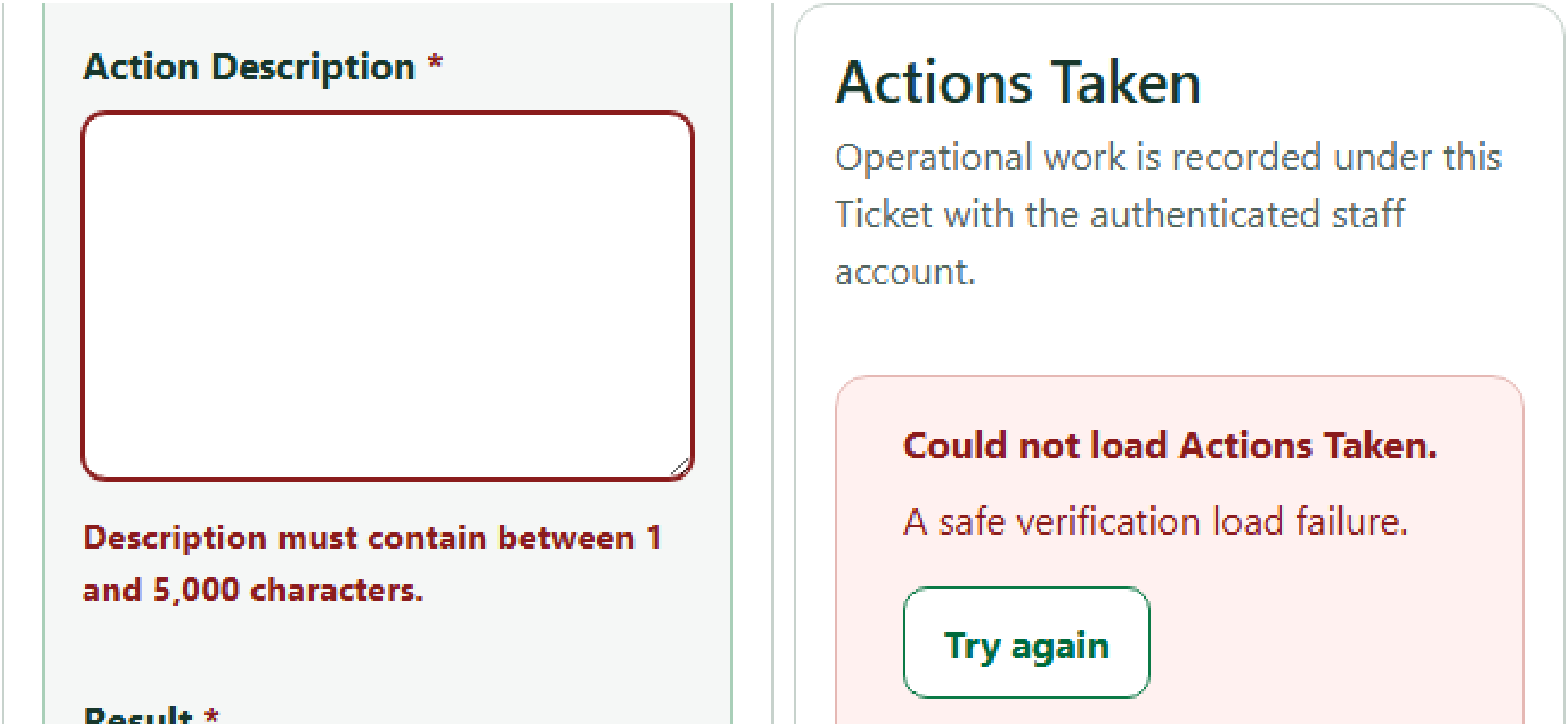
Desktop metric grid uses four columns, tablet at <=991px two, and mobile at <=767px one; Dashboard lists/breakdowns and two-column form grids stack at <=767px. Action text wraps; headers stack on narrow screens. Vertical scrolling is normal: a viewport screenshot ending at the fold does not prove a missing/clipped below-fold control. Full editor element captures show all fields and Save/Discard even when the form exceeds viewport height. Single-line readonly input values can scroll internally; the label and control must remain accessible and inside the viewport.

Check	Executed main evidence / scope
All major Lab 4 screens / three widths	Staff/Requester Dashboard, Action readonly/list/create/edit; full page and element captures at 1440x1000, 834x1112, 390x844
Feedback / validation	Dashboard success/empty/loading/forbidden/failure; live Action validation; controlled conflict/refresh failure/forbidden. Each HTTP fixture labelled separately
Readonly / role	Performer readonly; Requester Add/Edit absent; forbidden hides protected data/editor
Keyboard / focus	6 RESP-04 flows: Tab, Enter, Space, focus-visible; dashboard drill-down and editor checkbox/note/Save
Horizontal fit	Every supplementary capture asserts document scrollWidth <= clientWidth and all controls within bounds
Visual consistency / clipping / overlap	All gallery images manually inspected, including full-page below-fold content; existing Lab 3 captures reviewed for shell/navigation/private-content consistency
Browser hygiene	6 E2E browser scans: 0 unexpected HTTP failures/page errors; expected auth/me 401 probes separately counted
Limits	Targeted keyboard/semantic/visual checks, not full WCAG/axe certification or measured contrast audit; physical attachment recovery not proved by DB dump

Final peer-review gate: evidence PR approval/merge and accepted final submission remain unchecked. Then capture all Issues Done, record final evidence commit and publish the accepted PDF. The repository/main branch remains the source of truth.

Mobile validation and retained draft controls - readable regions

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Selected readable image regions; full originals are linked in the gallery. mobile-actions-validation.png / mobile-actions-refresh-failure-draft.png

Left: real server rejects whitespace Description with field-linked validation. Right: controlled refresh 500 leaves submitted edit text intact; retry and explicit discard remain visible. Full originals and provenance are clickable in the gallery.